

I. Single choice questions

1. The periodic signal with the smallest fundamental period among the following sequence is (**B**). 哪个最小基本周期的周期函数

A. $y[n] = 3 \cos(1.3\pi n) - 4 \sin(0.5\pi n + 0.5\pi)$

B. $y[n] = \cos\left(\frac{11\pi}{3}n\right)$

C. $y[n] = e^{j(\frac{n}{8}-\pi)}$

D. $y[n] = \cos\left(\frac{3\pi}{7}n - \frac{\pi}{8}\right)$

▲2. Let $x[n]$ be a real sequence, and $X(e^{j\omega})$ is the Discrete-time Fourier transform of $x[n]$. Which one of the following statements is wrong? (**B**) 实序列，DTFT，错误

A. The real part of $X(e^{j\omega})$ is an even function.

B. The image part of $X(e^{j\omega})$ is an even function.

C. $X(e^{j\omega})$ is a sequence with conjugate symmetry.

D. $\arg\{X(e^{j\omega})\} = -\arg\{X(e^{-j\omega})\}$.

✓ 1. 先讲通用结论：时域 $x[n]$ 是实数时，频域 $X(e^{j\omega})$ 有哪些性质？

不管时域是偶是奇，只要它是实的，就有：

属性	表达式	奇偶性
共轭对称性	$X(e^{j\omega}) = X^*(e^{-j\omega})$	-
实部	$\text{Re}\{X(e^{j\omega})\} = \text{Re}\{X(e^{-j\omega})\}$	偶函数 ✓
虚部	$\text{Im}\{X(e^{j\omega})\} = -\text{Im}\{X(e^{-j\omega})\}$	奇函数 ✓

✓ 2. 你问的“实偶/实奇”→“实偶/虚奇”是更具体情况，其实是在讨论如下变换结构：

我们把一个实序列 $x[n]$ 分为偶函数和奇函数部分：

$$x[n] = x_e[n] + x_o[n]$$

其中：

- 偶函数部分： $x_e[n] = \frac{1}{2}(x[n] + x[-n])$
- 奇函数部分： $x_o[n] = \frac{1}{2}(x[n] - x[-n])$

✦ 对应的频域变换会是：

时域	频域 DTFT
实偶函数 $x[n] = x[-n]$ 且实	DTFT 是 实偶函数 (频域实部有值，虚部为 0)
实奇函数 $x[n] = -x[-n]$ 且实	DTFT 是 纯虚奇函数 (频域虚部有值，实部为 0)

3. $y[n]=x[-n-1]$ is characterized by (B).

A. linear & causal

B. linear & non-causal

C. non-linear & causal

D. non-linear & non-causal

▲4. Let $x(n)=\delta(n)$, and its N-point DFT $\{x(n)\}=X(k)$, then $X(N-1)=$ (B).

A. N-1

B. 1

C. 0

D. 1-N

定义式不要忘记了:

$$X(k) = \sum_{n=0}^{N-1} x(n) \cdot e^{-j\frac{2\pi}{N}kn}$$

5. The order in which the input sequence $x[n]$ with $N=8$ in the decimation-in-time

FFT algorithm is stored in the storage unit should be (D).

A. $x[7]Xx[6]Xx[5]Xx[4]Xx[3]Xx[2]Xx[1]Xx[0]$;

B. $x[0]Xx[2]Xx[4]Xx[6]Xx[1]Xx[3]Xx[5]Xx[7]$

C. $x[4]Xx[5]Xx[6]Xx[7]Xx[0]Xx[1]Xx[2]Xx[3]$

D. $x[0]Xx[4]Xx[2]Xx[6]Xx[1]Xx[5]Xx[3]Xx[7]$

6. Among the following IIR filter structures, the structure that is most sensitive to the change of coefficient is (A). IIR 的物理结构有

A. Direct form

B. Cascade form

C. Parallel I form

D. Parallel II form

7. The difference equation of the filter $H(z) = \frac{z}{(2z-1)(z+0.6)}$ is (C).

A. $y[n] = 2x[n] + 0.2x[n-1] - 0.6x[n-2]$

B. $y[n-1] = 2x[n] + 0.2x[n-1] - 0.6x[n-2]$

C. $x[n-1] = 2y[n] + 0.2y[n-1] - 0.6y[n-2]$

D. $x[n] = 2y[n+2] + 0.2y[n+1] - 0.6y[n]$

被坑了!! 差分性质搞反了

重新写 $H(z) = \frac{z}{(2z-1)(z+0.6)}$, 两边同时乘以分母:

$$Y(z)(2z^2 + 0.2z - 0.6) = zX(z) \Rightarrow X(z) = \frac{Y(z)(2z^2 + 0.2z - 0.6)}{z}$$

再整理:

$$X(z) = Y(z)(2z + 0.2 - 0.6z^{-1})$$

转为时域:

- $Y(z) \cdot z \rightarrow y[n-1]$
- $Y(z) \rightarrow y[n]$
- $Y(z) \cdot z^{-1} \rightarrow y[n+1]$

所以有:

$$x[n] = 2y[n-1] + 0.2y[n] - 0.6y[n+1] \Rightarrow x[n-1] = 2y[n-2] + 0.2y[n-1] - 0.6y[n]$$

回对选项:

- ☒ C. $x[n-1] = 2y[n] + 0.2y[n-1] - 0.6y[n-2]$

8. The filter with linear phase represented by the finite length unit impulse response below is (A). 这个需要考虑四类 FIR, 对称和长度的奇偶

- A. $h[n] = \delta[n] + 2\delta[n-1] + \delta[n-2]$ B. $h[n] = \delta[n] + 2\delta[n-1] + 2\delta[n-2]$
C. $h[n] = \delta[n] + 2\delta[n-1] - \delta[n-2]$ D. $h[n] = \delta[n] + 2\delta[n-1] - 3\delta[n-2]$

9. Which one is incorrect in the following descriptions of the real coefficient all-pass transfer function. (A) 选错误的

- A. If there is a zero at λ , then there is a zero at $1/\lambda^*$.
B. If there is a zero at λ , then there is a pole at $1/\lambda^*$.
C. The real coefficient all-pass transfer function is a bounded real (BR) transfer function.
D. $|A(e^{j\omega})|^2 = A(z)A(z^{-1})|_{z=e^{j\omega}}$.

10. The amplitude response of which analog prototype filters is equiripple in stopband and monotonical in passband. (C)

- A. Butterworth B. Type 1 Chebyshev
C. Type 2 Chebyshev D. Elliptic (Cauer)

II. Blank-fill questions

1. The signal $x(n) = e^{j0.2\pi n} + e^{-j0.2\pi n}$ is obtained by sampling at 8000Hz.

The real frequency of this signal is 800Hz

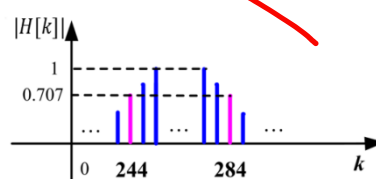
2. The transmission function of type I real coefficient FIR filter has the following zeros $z_1 = 1$, $z_2 = 0.6 + j0.8$. And the number of the remaining zeros of $H(z)$ with the lowest order is 2. They are 1, 0.6 - j0.8.

第一类没有固定特殊零点，而且一定有偶数个零点。且遵循“实”“偶”等关于零点关系的特征总结。

3. The difference equation of a IIR filter is $y[n] = \sum_{k=0}^M p_k x[n-k] - \sum_{l=1}^N d_l y[n-l]$. The direct-form II realization of this filter need $\max\{M, N\}$ delays, it's a canonic (canonic / noncanonic) structure.

4. Let the nonzero interval of sequence $x[n]$ and $y[n]$ be $[0, N-1]$ and $[0, M-1]$ respectively. Perform L-point circular convolution on $x[n]$ and $y[n]$. When L satisfy $L \geq M+N-1$, the circular convolution has the same result as their linear convolution. 其实卷积长度就是两个有一个 point 叠起来的时候大于这个就行

5. The amplitude frequency response of DFT at $N=1024$ points in a certain system $|H[k]|$ ($0 \leq k \leq N/2$) is shown in the following figure. And its amplitude spectrum decays by 3dB at $k=244$ and $k=284$. Let sampling frequency $f_s = 1024\text{kHz}$, the 3dB bandwidth of the system is 40 kHz. 这道题的痛点在于看不懂题。



核心公式

对于幅度比 (Amplitude Ratio) :

$$\text{dB} = 20 \cdot \log_{10} \left(\frac{A_2}{A_1} \right)$$

系统的频率响应 $|H[k]|$ 在第 $k = 244$ 和 $k = 284$ 这两个点时，下降到了最大值的 0.707 (也就是下降了 3dB)。

当 $\frac{A_2}{A_1} = 0.707$ 时:

$$20 \cdot \log_{10}(0.707) \approx -3.01 \text{ dB}$$

又告诉我们系统使用的 DFT 是 1024 点的，采样频率是 $f_s = 1024 \text{ kHz}$ 。

所以我们称:

那么问题是: 从频率响应的角度看，这个系统的“3dB 带宽”是多少? 当一个系统的频率响应下降到最大值的 0.707 倍时，衰减了 3dB。

📖 这题怎么计算?

DFT 有 1024 个点, 对应的采样频率是 1024kHz,
所以每一个频点代表的频率就是:

$$\frac{f_s}{N} = \frac{1024 \text{ kHz}}{1024} = 1 \text{ kHz}$$

也就是说:

- 第 244 点对应 244kHz
- 第 284 点对应 284kHz
- 所以带宽就是:

$$284 - 244 = \boxed{40 \text{ kHz}}$$

✅ 总结一句话:

题目图上告诉你: “从第244点到284点之间是下降了3dB”,
而每一点代表 1kHz, 所以频率间隔就是 40kHz, 这就是**带宽**。

6. The unit impulse response of system 1 is $x[n]$, which length is 16. The unit impulse response of system 2 is $h[n]$, which length is 32. Then the unit impulse response of a new system constructed by cascading the above two systems is $x[n]*h[n]$; its length is 47; The new system is stable (stable / unstable).

III. Calculations

1. (10 points) $x[n]$ and $y[n]$ are conjugate symmetric and conjugate

antisymmetric sequence respectively. Determine the symmetry of $x[n]*y[n]$.

Answer:

$$x^*[-n] = x[n], y^*[-n] = -y[n]$$

(2points)

let $w[n] = x[n]*y[n] = \sum_{k=-\infty}^{+\infty} x[k]y[n-k]$

(3points)

then $w[-n] = \sum_{k=-\infty}^{+\infty} x[k]y[-n-k]$

(1points)

$$\begin{aligned} w^*[-n] &= \sum_{k=-\infty}^{+\infty} x^*[k]y^*[-n-k] \\ &\stackrel{k=-l}{=} \sum_{k=-\infty}^{+\infty} x^*[-l]y^*[-n-l] \\ &= - \sum_{l=-\infty}^{+\infty} x[l]y[n-l] \\ \therefore x[n]*y[n] &\text{ antisymmetric sequence} \end{aligned}$$

把这个共轭的符号消掉了! ? (4points)

2. Let the DFT of the 8-length sequence $\{v[n]\}$ be $\{V[k]\} = \{-2+j6, 3+j5, 6+j4, -1+j8, -2+j3, 7+j3, -j8, 4-j7\}$. Calculate the DFT of the real part $\{x[n]\}$ and image part $\{y[n]\}$ of the sequence $\{v[n]\}$. 因为题目说了要分虚实来求 DFT，根据对称性的知识，时域的虚实和 DFT 后的共轭相对应，所以先直接把已知的 $V[K]$ 分解为 CS 和 CA 两个部分，然后进行计算。主要就是对称性转换这个地方。

Answer $X[k] = \frac{1}{2}\{V[k] + V^*[\langle -k \rangle_8]\}$, $Y[k] = \frac{1}{2j}\{V[k] - V^*[\langle -k \rangle_8]\}$ (4points)

$$V[k] = [-2 + j6, 3 + j5, 6 + j4, -1 + j8, -2 + j3, 7 + j3, -j8, 4 - j7],$$

$$V^*[\langle -k \rangle_8] = [-2 - j6, 4 + j7, j8, 7 - j3, -2 - j3, -1 - j8, 6 - j4, 3 - j5]. \quad (3points)$$

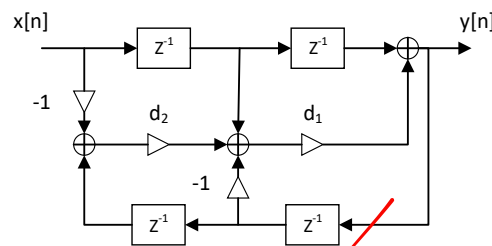
$$X[k] = \frac{1}{2}\{V[k] + V^*[\langle -k \rangle_8]\}$$

$$= [-2, 3.5 + j6, 4 + j6, 3 + j2.5, -2, 3 - j2.5, 3 - j6, 3.5 - j6],$$

$$Y[k] = \frac{1}{2j}\{V[k] - V^*[\langle -k \rangle_8]\}$$

$$= [6, -1 + j0.5, -2 - j3, 5.5 + j4, 3, 5.5 - j4, -2 + j3, -1 - j0.5]. \quad (3points)$$

3. Find the transfer function of the filter shown in the following figure.



Answer: Let the adder output from left to right be U_1, U_2, U_3

$$\begin{cases} U_1 = -X + Yz^{-2} & (1) \\ U_2 = Xz^{-1} + d_2U_1 - Yz^{-1} & (2) \\ Y = d_1U_2 + Xz^{-2} & (3) \end{cases} \quad (4points)$$

$$U_2 = Xz^{-1} - d_2X + d_2Yz^{-2} - Yz^{-1}$$

$$Y = d_1Xz^{-1} - d_1d_2X + d_1d_2Yz^{-2} - d_1Yz^{-1} + Xz^{-2}$$

$$Y(1 + d_1z^{-1} - d_1d_2z^{-2}) = X(-d_1d_2 + d_1z^{-1} + z^{-2}) \quad (2points)$$

$$H = \frac{-d_1d_2 + d_1z^{-1} + z^{-2}}{1 + d_1z^{-1} - d_1d_2z^{-2}} \quad (2points)$$

4. The impulse response of causal LTI discrete-time systems

is $h[n] = 2.5(0.5)^n\mu[n] + 0.5(-0.1)^n\mu[n]$

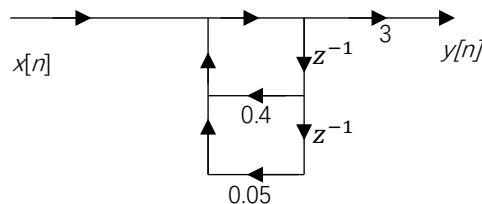
(1) Find the transfer function of the system.

(2) Give the canonical direct-form structure of the system.

Answer:

(1) $H(z) = \frac{2.5}{1-0.5z^{-1}} + \frac{0.5}{1+0.1z^{-1}} = \frac{3}{1-0.4z^{-1}-0.05z^{-2}}, |z|>0.5$ 4points

(2) $H(z) = \frac{3}{1-0.4z^{-1}-0.05z^{-2}}$ 2points



4points

5. (10 points) The transfer function of an analog prototype filter is

$$H_a(s) = \frac{s-2}{s+2}$$

(1) Try to obtain the transfer function of the digital IIR filter using bilinear transformation ($T=2$);

(2) Is the digital IIR filter stable? What kind of transfer function does the transfer function of the filter belong to? 判断稳定性的方式: 极点在单位圆内。最大相位系统是指它的极点 (poles) 都位于单位圆内, 且所有的零点都位于单位圆外。

Answer:

(1) $s = \frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}} = \frac{1-z^{-1}}{1+z^{-1}}$ 3points

$H(z) = H_a(s) \Big|_{s=\frac{1-z^{-1}}{1+z^{-1}}} = -\frac{1+3z^{-1}}{3+z^{-1}}$ 3points

(2) $H(z)$ pole at $-1/3$, So it's stable. 2points

It's an allpass system or maximum phase system. 2points

6. The parameters of linear phase FIR bandpass filter are: sampling frequency 20kHz, center frequency 4kHz, passband cutoff frequency at 3.5k and 4.5kHz, transition band

width 400Hz, $\alpha_s = -40dB$. Design the filter using the window function given in the appendix table, and the order of the filter is required to be as small as possible. Write out the unit impulse response of the filter.死记硬背这个过程就可以了

Answer:

(1): 这个采样公式又出现了这个地方有细节, 左边减去 400 赫兹右边加上 400 赫兹。

$$\begin{aligned}\omega_{p1} &= 2\pi \frac{3.5k}{20k} = 0.35\pi, & \omega_{p2} &= 2\pi \frac{4.5k}{20k} = 0.45\pi \\ \omega_{s1} &= 2\pi \frac{3.1k}{20k} = 0.31\pi, & \omega_{s2} &= 2\pi \frac{4.9k}{20k} = 0.49\pi\end{aligned}$$

----- (2points)



(2) According $\alpha_s = -40dB$, We choose hanning window 下面有表格我们只需要确定序列长度, 公式也给了

$$M = \frac{3.11\pi}{2\pi BW / f_s} = \frac{3.11 * 20000}{400 * 2} = 77.75 \approx 78$$

$$w[n] = 0.54 + 0.46 \cos\left(\frac{2\pi n}{2M+1}\right) = 0.54 + 0.46 \cos\left(\frac{\pi n}{157}\right) \quad -78 \leq n \leq 78 \text{ ----- (2points)}$$

$$(3) \quad \omega_{c1} = \frac{\omega_{p1} + \omega_{s1}}{2} = 0.33\pi, \quad \omega_{c2} = \frac{\omega_{p2} + \omega_{s2}}{2} = 0.47\pi \text{ ----- (2points)}$$

$$h_{bp}[n] = \frac{\sin(\omega_{c2}n)}{\pi n} - \frac{\sin(\omega_{c1}n)}{\pi n} = \frac{\sin(0.47\pi n)}{\pi n} - \frac{\sin(0.33\pi n)}{\pi n} \text{ ----- (2points)}$$

$$(4) \quad h_t[n] = h_{bp}[n]w[n] = \left[\frac{\sin(0.47\pi n)}{\pi n} - \frac{\sin(0.33\pi n)}{\pi n} \right] w[n] \quad -78 \leq n \leq 78 \text{ - (2points)}$$

$$(5) \quad h_d[n] = h_t[n-78] \quad 0 \leq n \leq 157 \text{ ----- (2points)}$$

Appendix:

Table1 Some fixed window functions

Rectangular:	$w[n] = \begin{cases} 1 & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$
hann:	$w[n] = \frac{1}{2} \left[1 + \cos\left(\frac{2\pi n}{2M+1}\right) \right], -M \leq n \leq M$

hamming:	$w[n] = 0.54 + 0.46 \cos\left(\frac{2\pi n}{2M+1}\right), -M \leq n \leq M$
blackman:	$w[n] = 0.42 + 0.5 \cos\left(\frac{2\pi n}{2M+1}\right) + 0.08 \cos\left(\frac{4\pi n}{2M+1}\right), -M \leq n \leq M$

Table 2 Properties of some fixed window functions

Type of window	Main lobe width Δ_{ML}	Relative Sidelobe Level A_{sl}	Minimum Stopband Attenuation	Transition Bandwidth $\Delta\omega$
Rectangular	$4\pi/(2M+1)$	13.3dB	20.9dB	$0.92\pi/M$
Hann	$8\pi/(2M+1)$	31.5dB	43.9dB	$3.11\pi/M$
Hamming	$8\pi/(2M+1)$	42.7dB	54.5dB	$3.32\pi/M$
Blackman	$12\pi/(2M+1)$	58.1dB	75.3dB	$5.56\pi/M$